Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 32. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the test accuracy during training.

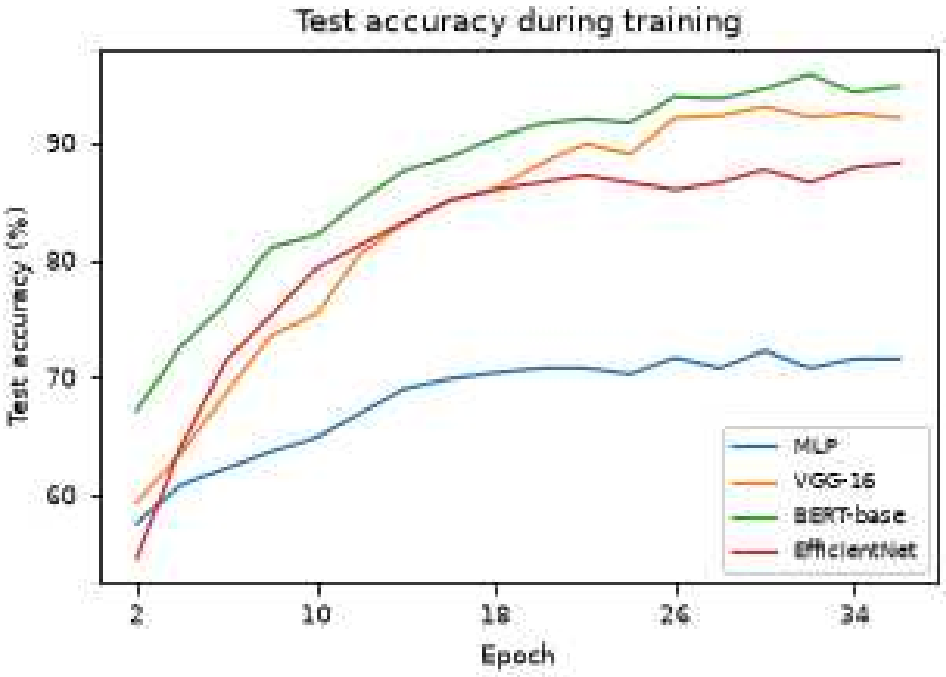


Figure 1: Test accuracy during training for the models in this experiment.

EfficientNet reaches 87.4% test accuracy at epoch 36. At epoch 28, BERT-base is 5.9 points better than VGG-16.

Figure 2 shows the training loss.

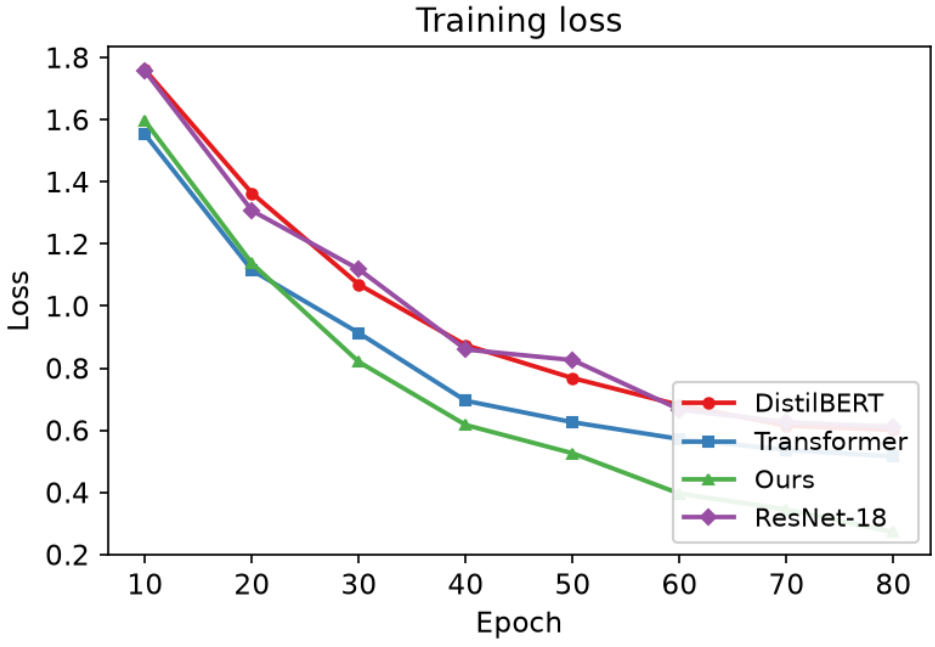


Figure 2: Training loss for the models in this experiment.

As shown in Figure 2, DistilBERT has a loss of 0.626 at epoch 70. DistilBERT has a loss of 1.095 at epoch 50. At epoch 20, the loss of Transformer is 1.115. As shown in Figure 2, at epoch 30, the loss of ResNet-18 is 1.118.

3 Discussion

Naive Bayes reaches 85.4% accuracy on SVHN. The differences between the models are discussed in the next section.